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SUPERSEDING
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MILITARY SPECIFICATION

# DYNAMOMETER, ENGINE RUN-IN, 1000 HORSEPOWER CAPACITY

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of The Department of Defense.

## 1. SCOPE

* 1.1 Scope. This specification covers a dynamometer designed and constructed to facilitate the testing and run-in of reciprocating-piston, internal combustion engines having outputs up to and including 1,000 brake horsepower (hp).

## 2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| O-A-548 | - | Antifreeze/Coolant, Engine, Ethylene Glycol, Inhibited, Concentrated |
| CC-M-1807 | - | Motor, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller) |
| PPP-B-636 | - | Box, Shipping, Fiberboard |
| PPP-C-843 | - | Cushioning Material, Cellulosic |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-LEE-S (R), Rock Island, IL 61299-6000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-T-60 - Tape, Packaging, Waterproof

MILITARY

MIL-C-104 - Crate, Wood, Lumber and Plywood, Sheathed, Nailed and Bolted
MIL-P-116 - Preservation, Methods of
MIL-T-152 - Treatment, Moisture and Fungus Resistant, Of Communications, Electronic, and Associated Electrical Equipment
MIL-C-3774 - Crate, Wood, Open, 12,000 and 16,000 Pound Capacity
MIL-C-5501 - Cap and Plug, Protective, Dust and Moisture Seal, General Specification for
MIL-W-5845 - Wire, Electrical, Iron and Constantan, Thermocouple
MIL-M-7298 - Manual, Technical, Commercial Equipment
MIL-S-12514 - Starter, Motor; Across-the-Line and Reduced Voltage Types, A-C Induction, 3/4 to 100 HP, Manual and Electric
MIL-I-24092 - Insulating Varnish, Electrical, Impregnating, Solvent Containing

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards For Federal Services

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents form part of this specification to the extent specified herein.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CRF 1910).

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)



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* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION (NEMA)

Publication WC3 - Rubber Insulated Wire and Cable for the Transmission and Distribution of Electrical Energy
NEMA Electrical Standards

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 2101 L Street, NW, Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, Battery March Park, Quincy MA 02269.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete dynamometer for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

* 3.1.1 Certification. Certification, supported by test data/reports, that the dynamometer is capable of absorbing power as described on the power absorption curve (see 3.4.1.1) shall be furnished with the first article or test report, as applicable. The power curve shall also be included in any technical publications furnished with each dynamometer.



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* 3.2 Design. The dynamometer shall be designed for the mounting, testing, and run-in of reciprocating piston, spark ignition and compression ignition, internal combustion engines up to and including 1,000 horsepower at speeds from 400 to 4,000 revolutions per minute (rpm). The dynamometer shall be capable of testing and run-in of the following engines: Continental AV-1790-5 through 5B and -7 through 7C, AVI-1790-8, AVSI-1790-6 and 6A, AVDS-1790-2 and 2A, AOSI-895-5, AOS-895-3, AO-895-4, AOI-402-5, and Ford GAA. The engine dynamometer shall be so designed that the engines may be readily installed with only minor adjustment for the particular engine being tested. The dynamometer shall include all the equipment necessary, including engine adapters, for mounting and proper operation of the above engines, except for the following, which (if required) shall be furnished by the Government:

(a) Coolant supply
(b) Coolant disposal
(c) Compressed air supply
(d) Electric power supply
(e) Engine fuel supply
(f) Engine exhaust disposal
(g) Engine hoisting equipment
(h) Fire extinguishing equipment
(i) Instruments for measuring atmospheric conditions
(j) Storage batteries

* 3.2.1 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings, and filler caps shall be accessible without requiring disassembly of functional parts. Lubrication reservoirs, if applicable, shall have means for checking levels.

* 3.2.2 Safety and health requirements. Safety devices shall be provided for all parts presenting safety hazards which can be guarded without substantial interference with operation of the dynamometer. All guards shall be removable to provide easy access to guarded parts. The dynamometer shall comply with the general safety requirements promulgated under OSHA 2206 applicable to the dynamometer itself. The dynamometer shall also comply with the National Electrical Code, NFPA 70. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.3).

* 3.2.2.1 Overspeed trip governor. An engine overspeed trip governor shall be furnished to provide overspeed protection for both spark ignition and compression ignition engines. The overspeed governor shall override the throttle control and shall either stop the engine completely or return the engine to a fast idle in the event the engine speed exceeds the governor setting. The overspeed governor shall be graduated in increments not greater than 100 rpm, and shall be adjustable through a 2,000 to 4,000 rpm engine speed range. The governor speed selector shall be panel mounted on the control console specified in 3.4.8.

* 3.2.3 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.



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* 3.3 Construction.

* 3.3.1 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the automotive equipment industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

* 3.3.1.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

* 3.3.2 Castings and forgings. All castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability.

* 3.3.3 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

* 3.3.4 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

* 3.3.5 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

* 3.3.6 Surfaces. All surfaces of the dynamometer shall be clean and free from any harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

* 3.3.7 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.4 Performance and product characteristics.



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* 3.4.1 Power absorption system. The power absorption system shall include a power absorber, power absorber frame, and load controls as specified in 3.4.1.1 through 3.4.1.3.

* 3.4.1.1 Power absorber. The power absorber (and complete dynamometer) shall be capable of continuously absorbing the horsepower values at the revolutions per minute (rpm) given below. All values shall be considered to be observed values:

330 hp at 1000 rpm ± 50 rpm
500 hp at 1,150 ± 50 rpm
800 hp at 1,700 ± 100 rpm
1,000 hp at 2,100 ± 100 rpm
1,000 hp at 4,000 ± 100 rpm

In the event an engine with the proper capacity cannot be obtained, it is permissible to utilize a transmission, gears, or other means to reduce speed or increase speed to obtain the horsepower and speed combination specified.

The minimum controllable hp curve shall not exceed the following:

12 hp at 1,000 ± 100 rpm
40 hp at 2,000 ± 100 rpm

The power absorber shall be capable of reproducing the same speed at any given horsepower, without readjusting the controls, after being stopped or speeded up. The power absorber shall be hydraulic, eddy current, or electric generator type and shall be of bi-directional rotation. Connections to the power absorber shall be such that they cannot affect power measurements. The power absorber shaft bearing seals for water cooled and hydraulic types shall be capable of maintaining a seal under normal operating conditions without requiring external cooling. The seals shall prevent cooling water and lubricant from leaking out and foreign material from entering. Seal replacement shall not require major dynamometer disassembly. The power absorber shaft bearings shall be of the antifriction type. When the power absorber is water cooled or hydraulic type, seals shall be provided to prevent fluid from entering the bearings. Adequate means of dissipation of absorbed power shall be provided and shall be in accordance with 3.4.3. Load control of the power absorber shall be in accordance with 3.4.1.3. Temperature control of the power absorber, if required for proper dynamometer operation, shall be in accordance with 3.4.3.

* 3.4.1.2 Power absorber frame. The power absorber shall be mounted in the power absorber frame. The bearings supporting the power absorber shall be antifriction type, and designed for at least 20,000 hours of operation without failure or degradation of dynamometer accuracy due to bearing race identation under static vibratory load. The frame shall support the power absorber and related equipment for all specified dynamometer operational requirements.

* 3.4.1.3 Load controls. The dynamometer shall be capable of controlling the load on the engine to within ± one percent of the torque indicator full scale value, and ± one percent of the full scale horsepower indicator value. The load controls shall provide coarse and fine adjustment of any load within the



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dynamometer capacity range in no more than one minute. The dynamometer shall maintain the selected load without manual manipulation of controls. Provision shall be made for the dynamometer to automatically return to preset load within 15 seconds after encountering temporary changes in engine speed and load caused by shorting out spark plugs, reducing throttle, or running an acceleration test. The power absorber load shall be controlled by controls mounted on the control console specified in 3.4.8.

* 3.4.2 Engine stands. Two mobile engine stands, capable of supporting engines weighing up to 10,000 pounds shall be provided and each stand shall be mounted on a minimum of four wheels. The wheels on one end shall be the rigid mounted, nonswivel type, and the remainder of the wheels shall be 360 degree swivel caster type. The wheels shall have a minimum diameter of four inches and shall provide free steerable movement of the loaded engine stand in any direction over the floor surface. The wheels shall be provided with antifriction bearings in the swivels and wheels. Wheel location on the engine stand shall effectively support movement and prevent an unbalanced or tipping condition of the stand when loaded with the maximum weight of 10,000 pounds. The swivel wheels shall be provided with foot actuated brakes, which may be applied to prevent movement of the engine stand. Mechanical guides shall be provided in conjunction with the engine stands to provide for proper engine installation and alignment (see 3.2). The stands shall be equipped with a drain pan having a tapered pipe thread of sufficient size (3/4 NPT minimum) with mating pipe cap. No hard plumbing is required.

* 3.4.3 Cooling system, engine, and dynamometer. A thermostatically controlled closed loop heat exchanger cooling system (cooling column, tower, radiator), shall be furnished for use with liquid cooled engines and the dynamometer. The heat dissipation rate shall be at least 5,100,000 BTU per hour in a maximum ambient air temperature of 120°F. The cooling system shall provide coolant to the operating engine at a temperature of 140 to 170°F. The cooling system is not to include any heating capability. A pressure relief valve shall be incorporated in the cooling system to prevent abnormal cooling system pressure. A means shall be provided for filling and draining the cooling system. Quick disconnect couplings shall be provided for connections between the cooling system and those provided on the engine stand. The cooling system shall be equipped with a sight gage to indicate the coolant level and a dial thermometer to indicate the engine coolant temperature. The system shall provide for manual control in the event of automatic control failure. A low coolant pressure safety switch shall be connected to the trip governor control (see 3.2.2.1) in such a manner as to reduce engine speed to a fast idle in the event of coolant system pressure failure. A cleanable, replaceable screen of 40 mesh shall be incorporated in the cooling system to screen out particles of rust and other foreign matter.



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3.4.4 Engine fuel system. The engine fuel system, supplied as part of the dynamometer system, shall conduct fuel from a supply outlet, provided by the Government (see 3.2), through a shutoff valve, fuel filters, selector valves, fuel flow measuring units, modulating valves, and fuel lines. The fuel system shall be incorporated with the function panels located on the junction box specified in 3.4.6. All valves shall be of the packless type, and all metal piping shall be brass or copper. The entire system shall permit passage of 850 pounds of fuel per hour with a pressure drop in the system of no more than 4 psig. Convenient and efficient bleed-offs shall be provided to eliminate vapor locks. Filters shall be of the replaceable element type and shall have a rating of not more than 5-microns. Two independent fuel systems, each with a primer pump, a shutoff and pressure modulating valve, and a pressure gage, shall be provided one for gasoline fueled carburetor and gasoline fuel injection type engines and one for compression-ignition type engines. Primers, pumps, valves, and pressure gages for both systems shall be provided.

* 3.4.5 Engine electrical system. The engine electrical system shall include provisions for two 12 volt vehicle wet storage batteries (wired in series) and an insulated wiring system to permit the operation of engines on negative grounded 12 and 24 volt ignition systems. The engine electrical system shall include provisions for remote control operation of degassers, starting relays, and booster coils through the use of an ignition switch mounted on the control console. Starting relays and circuits shall be capable of handling maximum starting currents. All cable used in the starting circuits shall be AWG copper conductor, heavy duty neoprene jacket, portable arc welding cable conforming to NEMA Publication WC3 and shall be capable of transmitting imposed electrical currents and voltages. An automatic battery charger, with a two stage charging rate, shall be equipped with an automatic regulator to provide a charging rate of 5 to 6 amperes when the battery voltage drops below normal, and to provide a "trickle" charge of not more than one ampere to keep the batteries fully charged at all times without the danger of over-charging or overheating the batteries. The charger shall be equipped with a meter and a meter switch to permit visual observation of the charging rate. The two 12 volt storage batteries shall not be furnished with the dynamometer.

3.4.6 Junction box. The junction box shall provide a compact area where all leads shall connect to their respective systems.

* 3.4.7 Control console leads. All electrical and mechanical leads required for remote engine operation shall be furnished, shall pass through separate conduits, and shall be grouped as follows:

(a) Instrument electrical leads
(b) Engine and operating 24 volt leads
(c) Thermocouple leads
(d) Primary power leads
(e) Mechanical leads

The length of the leads shall be 30 feet (minimum). Electrical leads shall terminate in terminal eyes. Instrument leads shall have soldered connections at both ends. All leads shall be plainly identified at both ends.



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3.4.8 Control console. The control console shall house all controls and instruments necessary for engine testing. The control console shall incorporate a data desk in a position below, and in front of, the instrumentation and control boards. Instruments, data disk, and controls shall be oriented to promote legibility and ease of operation by the seated operator. The instrumentation and control panel(s) above the desk level shall be arranged to permit a view of the engine dynamometer units. Access shall be provided for servicing and repairing all instruments and controls. The thermocouple and instrument circuits shall be incorporated with the junction (panels) on the junction box specified in 3.4.6. The control console shall be arranged for installation outside of the test cell. The instruments and controls shall be permanently marked for function and grouped, as far as possible, in four general areas as specified in 3.4.8.1, 3.4.8.2, 3.4.8.3, and 3.4.8.4.

* 3.4.8.1 Engine controls. The engine controls shall include the panel mounted ignition switch specified in 3.4.5, a panel mounted switch for the battery system, and the engine throttle control transmitter. Engine controls shall be included for the priming pumps for both fuel systems (see 3.4.4).

3.4.8.2 Engine instrumentation. The engine instrumentation shall include the following panel mounted indicators. The indicators shall be capable of receiving, transmitting, and indicating oil pressure, intake manifold pressure and vacuum, fuel pressure, fuel flow rate, cylinder head temperature, coolant temperature, and oil temperature as noted:

(a) One oil pressure indicating device. The pressure indicator shall be connected to the proper terminal on the junction panel (located on the junction box specified in 3.4.6).

(b) One mechanical engine oil pressure indicating device with a range from 0 to 160 pounds per square inch (psi), graduated in increments not greater than 5 psi, and with an accuracy error not greater than ± three percent of the full scale value from 0 to 40 psi and 120 to 160 psi and ± two percent of full scale value from 40 to 120 psi.

(c) Two mechanical fuel pressure indicating devices, one for gasoline and one for diesel fuel, with a range from 0 to 15 psig pressure and 0 to 30 inches of mercury vacuum, graduated in increments not greater than .5 psi and 1 inch, and an accuracy error not greater than ± five percent of full scale value.

(d) Two manifold pressure indicating devices with a range of 0 to 70 inches of mercury absolute pressure, graduated in increments not greater than .5 inch, and an accuracy error not greater than ± 1/2 inch of mercury.

(e) One flush mounted thermocouple type engine temperature indicator having at least eight stations with automatic cold junction temperature compensation, a range of 0 to 600°F, and an accuracy error not greater than ± one percent of full scale value. Eight pairs of No. 16 AWG, or larger, Iron-Constantan thermocouple lead wires in accordance with MIL-W-5845, type I, class E, shall be used from the instrument to the dynamometer junction panel thermocouple outlets. The instrument shall be compensated to match the length of lead wire from the indicating instrument to the engine thermocouples. A record receptacle space for recording the location on the engine of the thermocouple connected to each station shall be located near the temperature indicator selector switch. A supply of 100 record cards shall be furnished with each dynamometer.



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(f) Two fuel flow indicating devices with an accuracy error not greater than ± two percent of full scale value. Each fuel flow device shall be calibrated for gasoline with an API degree range of 52 to 62. One device shall have a minimum working range from 15 pounds per hour to 110 pounds per hour. The working range of the second device shall be from minimum 100 pounds per hour to 900 pounds per hour. A selector valve shall be provided to permit the use of either device for proper operation. The flow device and valves shall be located in a unit-constructed, airtight chamber equipped with a vent which shall conduct vapors to a non-hazardous area.

* 3.4.8.3 Dynamometer instrumentation and controls. The dynamometer instrumentation and control area shall include the following panel mounted indicators and shall be capable of receiving, transmitting, and indicating engine speed, torque, and horsepower, as noted. Digital display indicators may be substituted for the meters in (a) and (b) provided they perform the required functions, as noted, and have an accuracy error not greater than ± one percent of full scale value and ± one digit. All display devices shall be readable at a distance of five feet by the operator.

(a) One magnetic pulse type engine rpm sensor with leads to an electric tachometer indicating instrument. The indicating instrument shall have two scales. One scale shall have a range from 0 to 2,000 rpm graduated in increments not greater than 20 rpm, and the other scale shall have a range from 0 to 4,000 rpm graduated in increments not greater than 40 rpm. Each scale shall have a length of not less than 14 inches. The indicating instrument shall have an accuracy error not greater than ± 10 rpm.

(b) One torque and one horsepower measuring system with the torque sensing device directly actuated by the power absorber torque arm. All necessary leads shall be furnished from the torque sensing device to the torque and hp indicators. The torque and hp measuring systems shall have provisions for calibration (see also 3.4.11). The torque indicator shall have a range from 0 to 2,500 foot-pounds (minimum), graduated in increments not greater than 25 foot-pounds. The horsepower indicator shall have a range from 0 to 1,000 hp, graduated in increments not greater than 10 hp. The torque indicator shall have an accuracy error not greater than ± one percent of full scale value. The horsepower indicator shall have an accuracy error not greater than ± one percent of full scale value. The torque and horsepower indicating systems shall not require any temperature compensation control.

(c) One mechanical coolant supply pressure indicating device with a range of 0 to 100 psi graduated in increments not greater than 5 psi. Accuracy error shall be within ± three percent of full scale value.

(d) One mechanical power absorber fluid outlet temperature device (coolant and hydraulic type power absorbers only). Accuracy error shall be within ± three percent of full scale value.

(e) The dynamometer load controls shall be as specified in 3.4.1.3.



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* 3.4.8.4 Safety warning devices. Control console, panel mounted, individual warning lights or buzzers shall be furnished to indicate the following conditions:

(a) High power absorber fluid temperature (water-cooled and hydraulic type power absorbers only).
(b) Low engine oil pressure. The indicator light shall be connected to the proper terminal on the junction panel, located on the junction box specified in 3.4.6.
(c) High engine oil temperature. The indicating light shall be connected to the proper terminal on the junction panel, located on the junction box specified in 3.4.6.
(d) Oil from engine not circulating through oil system.
(e) Tripped overspeed governor.

* 3.4.9 Drive shaft. A dynamometer drive coupling shall be provided and shall be capable of transmitting power over the horsepower and speed ranges specified in 3.4.1.1. The drive coupling shall compensate for angular misalignment of at least three degrees between the power absorber shaft and the engine shaft. The drive coupling shall have a lengthwise adjustment of at least 1 1/2 inches.

* 3.4.10 Adapter kits. Adapter kits shall be provided for testing and run-in of the engines specified in 3.2. Painting of adapter kit components shall be in accordance with standard commercial practice provided the following minimum criteria are met or exceeded: all surfaces to be painted shall be cleaned and be free of all foreign matter detrimental to painting, and shall have at least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), applied.

3.4.11 Hardware and special tools. All hardware required for the complete and proper installation of the dynamometer shall be furnished. All special tools necessary for the proper installation, operation, and maintenance of the dynamometer shall be furnished, including equipment for calibrating the torque or horsepower measuring system.

3.4.12 Electrical equipment. Electrical components, including starters, relays, switches, wiring, and their enclosures, shall conform to, and be located in accordance with, NEMA Electrical Standards except as specified herein. No component shall be used in any application where the requirements exceed the manufacturer's rating for the part. Connections to screw type terminals shall be mechanically secured to prevent loss of tightness. Properly crimped or soldered (brazed) terminal lugs are acceptable. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in all soldering operations.

* 3.4.12.1 Motors and starters. Electric motors shall conform to the requirements of CC-M-1807. Starters shall conform to the requirements of MIL-S-12514.

3.4.12.2 Power source. The dynamometer shall be designed for operation on nominal 220/440 volt, 50/60 hertz (Hz), three-phase current. Control switches and circuits shall operate on a nominal 115-volt, 50/60 Hz, single-phase current.



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3.5 Fungus control. Electrical components and circuit elements shall be treated for fungus resistance in accordance with the type II requirements of MIL-T-152, except that the statement of treatment is not required. The windings of electric motors shall be given two impregnating coats of varnish conforming to the requirements of MIL-I-24092, grade CB, class 155 through class 220, composition I (manufacturer's option).

3.6 Product identification. The dynamometer shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

* 3.7 Repair Parts. Repair part identification in repair part manuals shall include the prime contractor's name, nomenclature, part number, and price. Repair part identification in repair part manuals shall also include the oiginal equipment/part manufacturer's (OEM) (actual part manufacturer) name, Federal Supply Code for Manufacturers (FSCM), nomenclature, part number, and price of the part used in the end item immediately after the prime contractor's part identification. Items with high mortality rates shall be so identified. (See MIL-M-7298.)

* 3.8 Workmanship. Standards of workmanship shall be such as to assure equipment with characteristics found in standard commercial units and as specified herein. Particular attention shall be given to freedom from blemishes, defects, and sharp edges; thoroughness of soldering, welding, brazing and painting; and alignment of parts and tightness of screws and bolts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When specified in the contract, first article inspection shall be applied to the preproduction model or initial production item (see 3.1). First article inspection shall consist of the examination in 4.5 and 4.6 and all tests under 4.7. Failure of the first article to pass the examinations or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production items offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (e) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).



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(b) Instrument accuracy (see 4.6).
(c) Power absorber test (see 4.7.1).
(d) Safety warning device test (see 4.7.2).
(e) Packaging inspection (see 4.8).

* 4.5 Product examination. Each complete dynamometer shall be examined to determine compliance with the requirements of Section 3. As a minimum, the examination shall be in accordance with the following checklist:

EXAMINATION CHECKLIST

| REQUIREMENT | PARAGRAPH |
|---|---|
| Design | 3.2 |
| Lubrication | 3.2.1 |
| Safety provisions | 3.2.2 |
| Overspeed trip governor | 3.2.2.1 |
| Interchangeability | 3.2.3 |
| Materials | 3.3.1 |
| Dissimilar metals | 3.3.1 |
| Castings and forgings | 3.3.2 |
| Welding, brazing, or soldering | 3.3.3 |
| Fastening devices | 3.3.4 |
| Threads | 3.3.5 |
| Surfaces | 3.3.6 |
| Painting | 3.3.7 |
| Power absorber construction | 3.4.1.1 |
| Power absorber frame construction | 3.4.1.2 |
| Mounting of load controls | 3.4.1.3 |
| Engine stand-guides and drain pan | 3.4.2 |
| Cooling system, engine, and dynamometer | 3.4.3 |
| Engine fuel system | 3.4.4 |
| Engine electrical system | 3.4.5 |
| Junction box | 3.4.6 |
| Control console leads | 3.4.7 |
| Control console | 3.4.8 |
| Engine controls | 3.4.8.1 |
| Engine instrumentation | 3.4.8.2 |
| Dynamometer instrumentation and controls | 3.4.8.3 |
| Safety warning devices | 3.4.8.4 |
| Drive shaft and coupling | 3.4.9 |
| Adapter kits | 3.4.10 |
| Hardware and special tools | 3.4.11 |
| Electrical equipment | 3.4.12 |
| Motors and starters | 3.4.12.1 |
| Fungus control | 3.5 |
| Product identification marking | 3.6 |
| Repair parts | 3.7 |
| Workmanship | 3.8 |

A certificate of conformance may be provided to satisfy the examination requirements where more than a visual or dimensional verification is necessary.



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* 4.6 Instrument accuracy. The indicating instruments specified in 3.4.8.2, 3.4.8.3, and 3.4.8.4 shall be verified for calibration accuracy, prior to initiation of acceptance testing of the dynamometer, by comparison with standard measuring devices having an overall accuracy error within 1/4 of one percent. Each instrument shall be inspected to determine conformance to the applicable requirements for accuracy.

4.7 Tests.

* 4.7.1 Power absorber test. The power absorber shall be motored with an engine at the following specified conditions:

(a) 330 hp at 1000 ± 50 rpm for one-half hour
(b) 500 hp at 1,150 ± 50 rpm for one-half hour
(c) 800 hp at 1,700 ± 100 rpm for two hours
(d) 1,000 hp at 2,100 ± 100 rpm for one hour
(e) 1,000 hp at 4,000 ± 100 rpm for one hour

In the event an engine with the proper capacity cannot be obtained, it is permissible to utilize a transmission, gears, or other means to reduce speed or increase speed to obtain the horsepower and speed combination specified.

The temperature of the power absorber shall not exceed that warranted by the manufacturer at each of the above specified load and speed conditions. Loss of load, loss of control, or temperature in excess of the manufacturer's warranty shall be cause for rejection of the dynamometer being tested.

* 4.7.2 Safety warning device test. The unsafe conditions of 3.4.8.4 shall be simulated by creating out-of-tolerance signal inputs to the sensing devices to determine that the safety warning devices warn of unsafe conditions. Failure of any warning device to operate as required shall be cause for rejection of the dynamometer being tested.

* 4.7.3 Operational test. The complete dynamometer shall be prepared for a test run in accordance with the manufacturer's instructions. A commercially available internal combustion engine, capable of meeting the test conditions specified below, shall be furnished by the contractor and used for this test. During the operational test, the indicating instruments of 3.4.8.2, 3.4.8.3, and 3.4.8.4 shall be tested to determine compliance with the performance and accuracy requirements specified herein.

Test Conditions

| Period | Time (Minutes) | Engine Speed (rpm) | Net Observed Brake Horsepower |
|---|---|---|---|
| 1 | 10 | 800 | No load |
| 2 | 10 | 1,000 | 12.0 (Minimum load control check) |
| 3 | 10 | 1,000 | 430.0 |
| 4 | 10 | 1,700 | 800.0 |
| 5 | 10 | 2,000 | 40.0 (Minimum load control check) |
| 6 | 10 | 2,000 | 940.0 |



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Test Conditions (Continued)

| Period | Time (Minutes) | Engine Speed (rpm) | Net Observed Brake Horsepower |
|---|---|---|---|
| 7 | 10 | Any speed between 2,100 and 4,000 | 1000.0 |
| 8 | 5 | 1,000 | No load |

Failure of any part of the dynamometer or any of its equipment to meet the applicable performance requirements shall be cause for rejection of the dynamometer being tested.

* 4.8 Packaging inspection. Packaging inspection shall be conducted before and after packaging the dynamometer to determine compliance with the requirements of Section 5.

5. PACKAGING

* 5.1 Disassembly. The dynamometer may be disassembled to the extent necessary to assure that all surfaces requiring a preservative are accessible for processing, to assure minimum cube for packing, and to permit reassembly without special skills or tools. Disassembly, when applicable, shall be as specified in MIL-STD-1186.

* 5.2 Preservation. Preservation shall be Level A or Industrial, as specified (see 6.2.1).

5.2.1 Level A.

5.2.1.1 Cleaning and drying. The dynamometer shall be cleaned in accordance with process C-1 and dried in accordance with procedures D-1 through D-5 of MIL-P-116.

5.2.1.2 Unit protection. The unit method of packaging and preservatives specified herein shall conform to MIL-P-116.

* 5.2.1.2.1 Electrical accessories and components. The control(s) and instrumentation panel, including all electrical accessories and components such as cables, motors, starters, junction boxes, electrical leads, and publications shall be packaged as a unit in accordance with the applicable sub-method of method II of MIL-P-116.

5.2.1.2.2 Power absorption unit. All exposed bare metal areas of the power absorption unit shall be coated with type P-11 preservative, and shall be packaged in accordance with method I. The power absorption unit may be packaged in accordance with method III when the absorption unit and frame are completely painted.

5.2.1.2.3 Oil and fuel systems. The oil and fuel systems, including storage tanks, shall be flushed with type P-10, grade 30 preservative. The systems shall be thoroughly drained. After draining, seal the inlets and outlets with tape conforming to PPP-T-60, type IV, or plastic caps or plugs conforming to MIL-C-5501.



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5.2.1.2.4 Water system. The complete water system shall be flushed with a solution consisting of 40 percent antifreeze (ethylene glycol) conforming to O-A-548, type I, and 60 percent clean water. After flushing, the system shall be thoroughly drained. All openings to the system shall be sealed with plastic caps or plugs conforming to MIL-C-5501 or tape conforming to PPP-T-60, type IV.

5.2.1.2.5 Bearings. All bearings used throughout the system and all revolving shafts shall be preserved with type P-11 preservative.

5.2.1.2.6 Assembly components. All assembly components such as engine stand, drive shaft, adapter set, pump assemblies, fastening devices (nuts, bolts, pins, washers, and screws), pipe, pipe fittings, and other hardware, including special tools, shall be packaged in accordance with method I of MIL-P-116. In the event the components are completely painted, plated, or fabricated of stainless steel, no preservative is required. When practicable, components shall be placed in fiberboard boxes conforming to PPP-B-636, class weather-resistant. Parts vulnerable to damage shall be cushioned with material conforming to PPP-C-843, type II.

5.2.1.2.7 Hose. All hose issued with the dynamometer shall be coiled to a safe diameter and secured with not less than three pieces of cotton twine or tape evenly spaced about the coil.

5.2.1.2.8 Gages and safety warning lights. Gages and safety warning lights shall be covered with one inch thick uncompressed cushioning material conforming to PPP-C-843, type II, or fiberboard conforming to PPP-F-320, class weather-resistant. Secure the material in place with tape conforming to PPP-T-60, type optional.

* 5.2.2 Industrial. Industrial preservation shall be as specified in ASTM D3951-82.

* 5.3 Packing. Packing shall be Level A or Industrial, as specified (see 6.2.1).

5.3.1 Level A. The packaged control console, complete with instrument panel and all electrical accessories, shall be packed in a crate conforming to MIL-C-104. All other component parts packaged shall be packed in one or more crates conforming either to MIL-C-104 or MIL-C-3774. Blocking, bracing, and anchoring shall conform to the crate specification and MIL-STD-1186. Items packed in open crates shall be covered with a shroud conforming to MIL-STD-1186. Closure and strapping shall conform to the crate specification.

* 5.3.2 Industrial. Industrial packing shall be as specified in ASTM D3951-82.

* 5.4 Marking. Marking shall be Level A or Industrial, as specified (see 6.2.1).

* 5.4.1 Level A. Level A marking shall be as specified in MIL-STD-129.

* 5.4.2 Industrial. Industrial marking shall be as specified in ASTM D3951-82.



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6. NOTES

6.1 Intended use. The dynamometer covered by this specification is intended to be used for testing and run-in of new or rebuilt reciprocating-piston, internal combustion engines prior to the release of the engines for use in vehicles or for forwarding to supply channels or storage facilities.

6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article, when required (see 3.1).
(c) Additional safety and health requirements (see 3.2.2 and 6.3).
(d) Painting, if different (see 3.3.7).
(e) Product identification marking, if different (see 3.6).
(f) Selection of level of preservation, packing, and marking required (see 5.2, 5.3, and 5.4).

6.3 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and controls of noise levels, radiation, electromagnetic emissions, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
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| | |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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